EE2004 Digital Signal Processing

Tutorial 6 Answer Key

1. (a) $X_1(\omega) = 1 + 2\cos\omega + 2\cos 2\omega$
   (b) $X_2(\omega) = 1 + 2\cos 2\omega + 2\cos 4\omega$
   (c) $X_3(\omega) = 1 + 2\cos 3\omega + 2\cos 6\omega$

2. FIR

3. (a) $4\,(1/2)^n\,u[n] - 2\,(1/4)^n\,u[n] - (n+1)\,(1/4)^n\,u[n]$
   (b) $\frac{2}{3}\,(-1)^n$

4. $H(e^{j\omega}) = \dfrac{1}{1 + \frac{1}{2}e^{-j\omega}}$

5. $3\left[\dfrac{\sin\frac{\pi}{2}(n-5)}{\pi(n-5)} - \dfrac{\sin\frac{\pi}{2}(n-6)}{\pi(n-6)}\right]$

6. (a) $H(e^{j\omega}) = \dfrac{\frac{4}{5}e^{-j\omega}}{1 - \frac{4}{5}e^{-j\omega}}$
   (b) $y[n] = \frac{4}{5}\,y[n-1] + \frac{4}{5}\,x[n-1]$

7. $y[n] = 0.4593\,\cos\left(\frac{n\pi}{8} + 1.5273\right) + 10.5263\,\cos\left(\frac{n\pi}{2}\right)$

8. (a) $y[n] = 2x[n] - x[n-1] - 0.125y[n-3]$
   (b) $2\,p[n] - p[n-1]$ where $p[n] = (-\frac{1}{8})^{n/3}\,u\left[\frac{n}{3}\right]$ whenever $n = 3k$ and zero otherwise.